FINMARK: FROM DATA TO DISCOVERY

Milestones 1 & 2 — Exploratory Data Analysis and Data Visualization

REWIND: WHERE OUR JOURNEY BEGAN

Why did we explore the data?

- Understand FinMark’s available customer data
- Identify data quality issues
- Discover customer, transaction, and engagement patterns
- Prepare for future machine learning development

Business Problem → Data Exploration → Insights → Future ML

MEET THE DATA: THREE DATASETS

Three datasets. One connected view of FinMark.

Customer Demographics

- 3,023 records
- Age, Gender, Location, Income Level, Signup Date

Customer Transactions

- 3,015 records
- Transaction Date, Amount, Product Category, Payment Method

Social Media Interactions

- 3,020 records
- Platform, Interaction Type, Sentiment

CustomerID: Connects records across datasets.

THE FIRST CHALLENGE: DATA QUALITY

What we found during data preparation:

- Duplicate rows were identified and removed
- Missing values were examined
- Data types and categories were reviewed
- Invalid and unusual values were flagged

Examples

- Age = −1
- Amount = −100, 0, and “Free”
- Missing values and “Unknown” categories

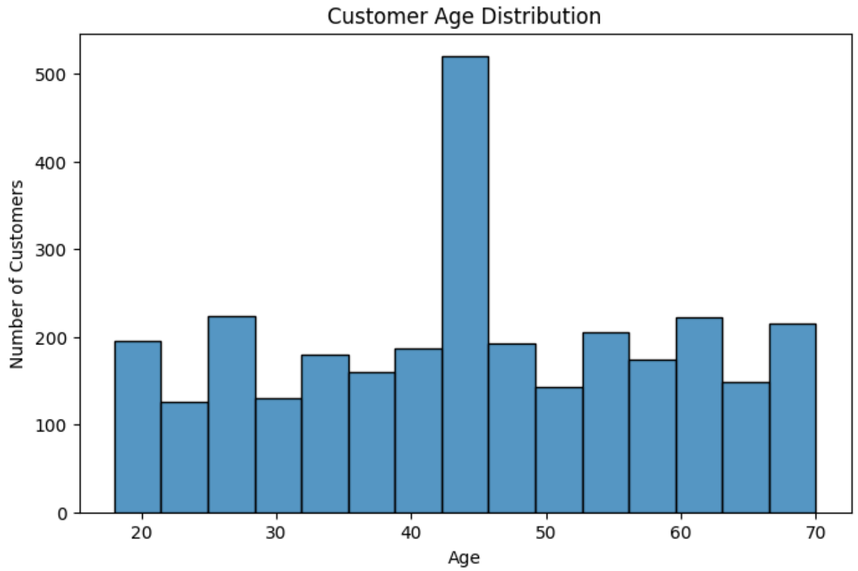
Raw Data → Data Preparation → Cleaned Data

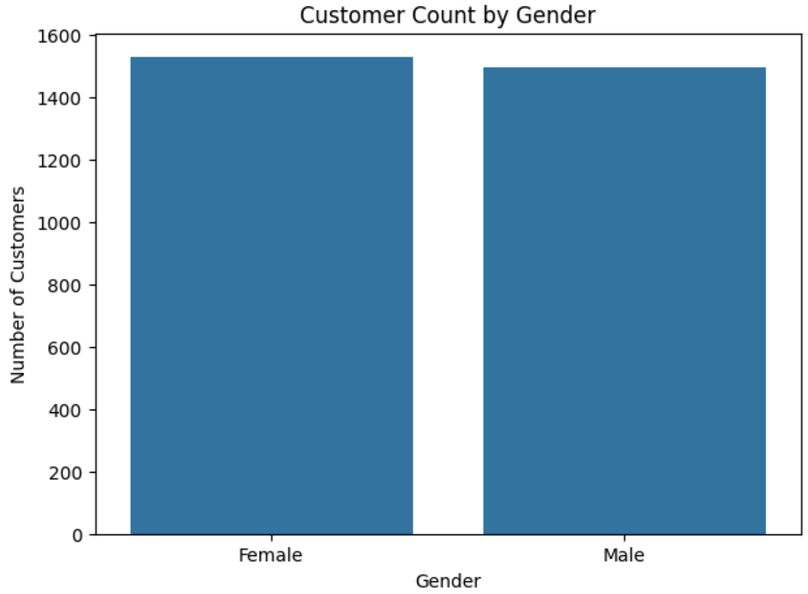
REWIND TO THE CUSTOMER PROFILE

Who are the customers in our data?

- Average age: **44.69 years**
- Median age: **45 years**
- Female: **1,528**
- Male: **1,495**

Key takeaway:
The recorded gender counts are nearly balanced.



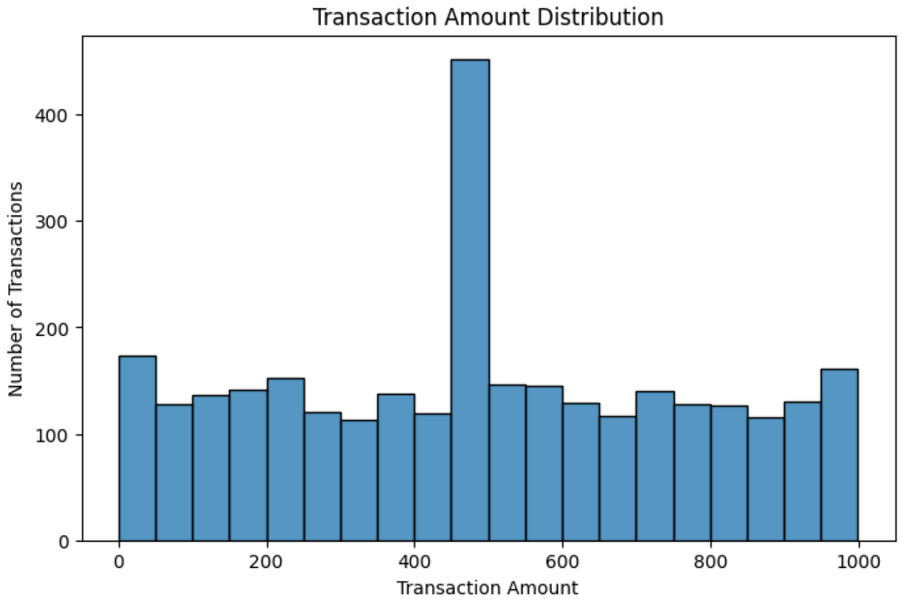


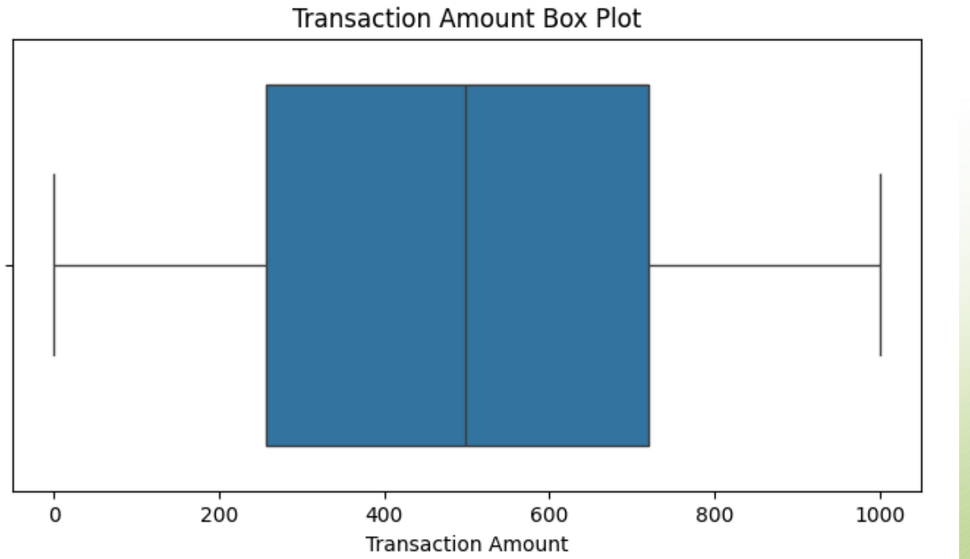
WHAT THE TRANSACTIONS REVEALED

Understanding recorded purchase amounts

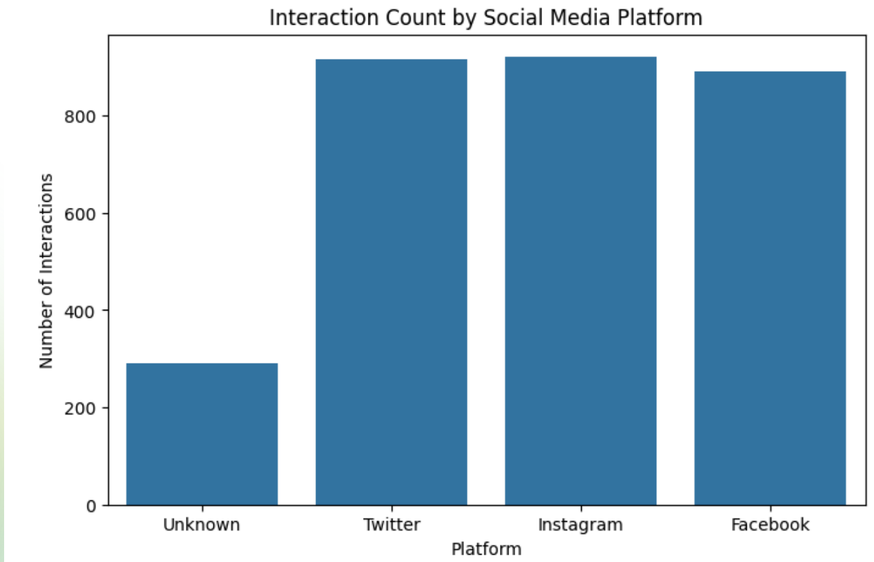
- Average amount: **493.62**
- Median amount: **498.015**
- Minimum: **0**
- Maximum: **999.86**
- Zero-value transactions: **60**
- Transactions with amount **498.015: 323**

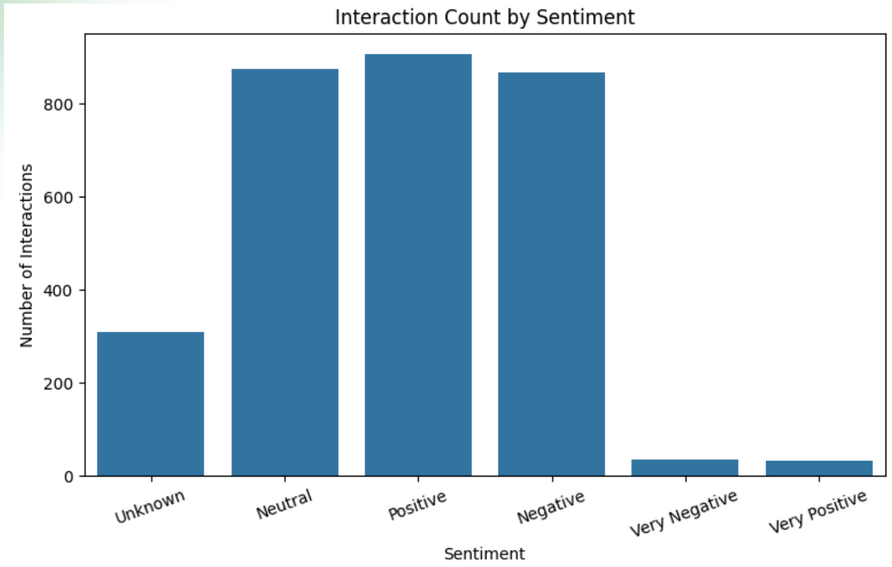
Key takeaway:
Zero and repeated amounts need further investigation.





LOOKING BACK AT SOCIAL ENGAGEMENT





What does the social data show?

Platforms

- Instagram
- Twitter
- Facebook
- Unknown

Interaction Types

- Comment
- Share
- Like

Sentiment Categories

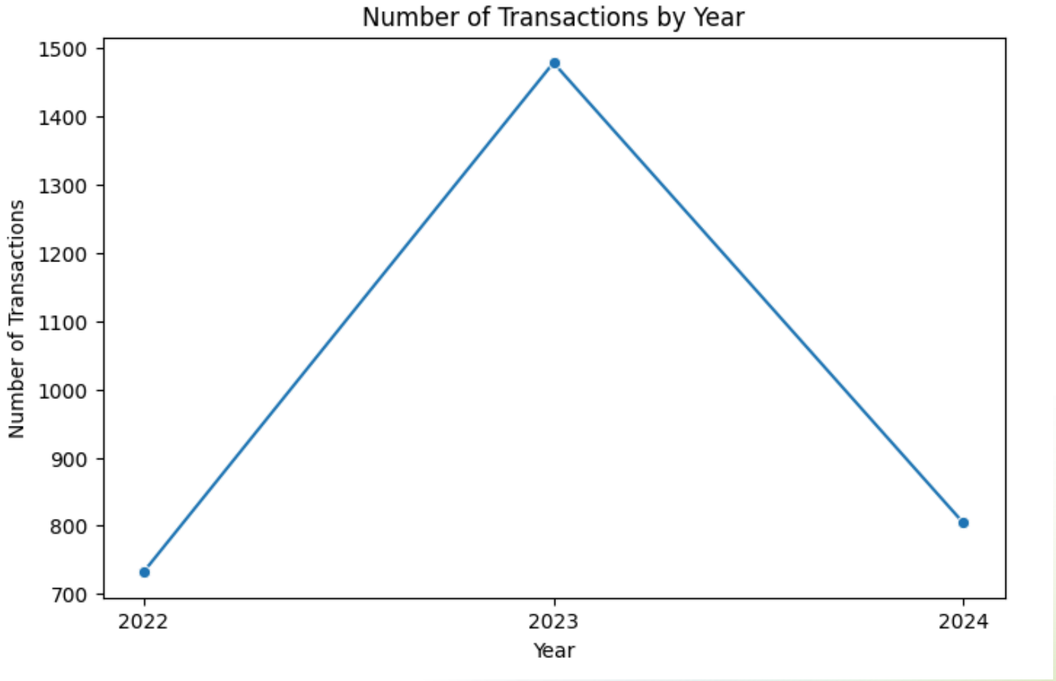
- Positive
- Neutral
- Negative
- Very Positive
- Very Negative
- Unknown

Key takeaway: *Engagement and sentiment vary across recorded categories.*

THE TIMELINE: PATTERNS AND LIMITATIONS

Transaction activity over time

- 2023 had the highest recorded transaction count
- Monthly data coverage is uneven
- Uneven coverage limits year-to-year comparisons
- More complete time data is needed for reliable trend analysis



WHAT WE LEARNED, WHAT COMES NEXT

What we learned

- What we learned
- The datasets provide complementary customer, transaction, and engagement information
- Data quality requires further validation
- Our visualizations are primarily descriptive
- CustomerID is useful for linking records, not automatically as a predictive feature

Next steps

- Investigate zero and repeated transaction amounts
- Clarify placeholder values such as “Unknown”
- Validate dates, IDs, and duplicate handling
- Explore relationships between variables
- Prepare for feature selection and model development

PROPOSED ML SOLUTION

From Data Insights to Predictive Modeling

Business Goal

Use FinMark's customer data to support better customer segmentation and more personalized marketing.

Proposed Approach

- Combine demographic, transaction, and social interaction data using CustomerID.
- Select relevant features such as age, income level, purchase amount, product category, and sentiment.
- Exclude identifiers such as CustomerID, TransactionID, and InteractionID from predictive features.
- Validate data quality and handle missing, unknown, and unusual values before modeling.

Model Development

- Compare suitable approaches, such as clustering for customer segmentation and classification if a reliable target label is available.
- Evaluate candidate models using appropriate metrics before considering deployment.

Current Status: EDA and data visualization completed. Model training and evaluation are future work.

From Data Exploration to Informed Decisions

Understand the data.
Validate its quality.
Interpret findings carefully.
Prepare for machine learning.

References

- Milestone 1: ML Solution EDA
- Milestone 2: ML Solution DataViz
- FinMark Customer Demographics Dataset
- FinMark Customer Transactions Dataset
- FinMark Social Media Interactions Dataset
- FinMark Week 7 Data Visualization Notebook

https://github.com/jrmhalzona/H3101-MLml-linear-algebra-project-Team-ML-4